

CD PE - 1

SRN : PES2UG23CS311

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Section : E

Code :

```
digit    [0-9]
letter   [_A-Za-z]
%{
#include<stdio.h>
%}
%%
auto|double|if|static|break|else|int|struct|case|enum|long|switch|char|extern|near|typedef|const|float|contin
ue|register|union|unsigned|void|while|default|do|goto|signed|return {printf("Keyword : \t %s\n",yytext);}

{letter}{letter}{digit}* {printf("Valid Identifier : %s\n",yytext);}

\<|>|=|<|>|<|> {printf("Rel_Op : %s\n",yytext);}

\&|\&|\||~ {printf("Log_Op : %s\n",yytext);}

\+|\-|\*|/ {printf("Arith_Op : %s\n",yytext);}

= {printf("Assign_Op : %s\n",yytext);}

\+|+ {printf("Inc_Op : %s\n",yytext);}

\|-|- {printf("Dec_Op : %s\n",yytext);}

\(|\)|\{|\}|\[|\]|;|, {printf("Punctuator : %s\n",yytext);}

{digit}+ {printf("Int constant : %s\n",yytext);}

{digit}+\.{digit}+ {printf("Floating pt constant : %s\n",yytext);}

\"(\|.|[^\"])*\" {printf("String literal : %s\n",yytext);}

'(\|.|[^\n])*' {printf("Char literal : %s\n",yytext);}

. ;

\n ;

%%
int yywrap()
{
```

```
    return(1);
}
int main(int argc, char *argv[])
{
    yyin = fopen(argv[1], "r");
    yylex();
    fclose(yyin);
    return 0;
}
```

Test code :

```
int add(int x, int y) {
    return x + y;
}

int main() {
    int i = 0;
    int sum = 0;

    while (i < MAX) {
        sum = sum + i;
        i++;
    }

    if (sum != 0 || i == MAX) {
        char ch = 'A';
        printf("Sum is %d\n", sum);
    }

    return 0;
}
```

Output screenshots :

```

Keyword :          int
Valid Identifier : add
Punctuator :(
Keyword :          int
Valid Identifier : x
Punctuator :,
Keyword :          int
Valid Identifier : y
Punctuator :)
Punctuator {:
Keyword :          return
Valid Identifier : x
Arith_Op :+
Valid Identifier : y
Punctuator ;;
Punctuator :}
Keyword :          int
Valid Identifier : main
Punctuator :(
Punctuator :)
Punctuator {:
Keyword :          int
Valid Identifier : i
Assign_Op :=
Int constant :0
Punctuator ;;
Keyword :          int
Valid Identifier : sum
Assign_Op :=
Int constant :0
Punctuator ;;
Keyword :          while
Punctuator :(
Valid Identifier : i
Rel_Op :<
Valid Identifier : MAX
Punctuator :)
Punctuator {:
Valid Identifier : sum
Assign_Op :=

```

```
Valid Identifier : sum
Assign_Op :=
Valid Identifier : sum
Arith_Op :+
Valid Identifier : i
Punctuator ;;
Valid Identifier : i
Inc_Op :++
Punctuator ;;
Punctuator :}
Keyword :      if
Punctuator :(
Valid Identifier : sum
Rel_Op :!=
Int constant :0
Log_Op :||
Valid Identifier : i
Rel_Op :==
Valid Identifier : MAX
Punctuator :)
Punctuator :{
Keyword :      char
Valid Identifier : ch
Assign_Op :=
Char literal : 'A'
Punctuator ;;
Valid Identifier : printf
Punctuator :(
String literal : "Sum is %d\n"
Punctuator :,
Valid Identifier : sum
Punctuator :)
Punctuator ;;
Punctuator :}
Keyword :      return
Int constant :0
Punctuator ;;
Punctuator :}
```